

Cardinal Preservation in Realizability for Set Theory

from a joint work with Laura Fontanella

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Realizability was conceived in 1945 by Kleene to formally relate the constructive approach motivating both intuitionism and recursion theory.

The very idea of this technique consists in assigning computationally meaningful witnesses for the validity of a formula. For instance

$$\begin{array}{lll} n \Vdash \varphi \wedge \psi & \text{iff} & \mathbf{p}_0 n \Vdash \varphi \wedge \mathbf{p}_1 n \Vdash \psi \\ n \Vdash \varphi \rightarrow \psi & \text{iff} & \forall m (m \Vdash \varphi \rightarrow \{n\}m \Vdash \psi) \\ n \Vdash \exists x \varphi(x) & \text{iff} & \mathbf{p}_1 n \Vdash \varphi[\mathbf{p}_0 n/x] \end{array}$$

This amounts to evaluating formulæ in an algebraic structure called combinatory algebra.

Kleene (1945) On the interpretation of intuitionistic number theory.
 Kleene (1973) Realizability: A retrospective survey.

A (partial) combinatory algebra $\mathcal{A}$ comprises a set A together with a (partial) binary operation $\cdot \bullet \cdot : A \times A \mapsto A$ and two fixed elements $\mathbf{k}, \mathbf{s} \in A$, such that

$$(\mathbf{k} \bullet a) \bullet b = a, \quad ((\mathbf{s} \bullet a) \bullet b) \bullet c = (a \bullet c) \bullet (b \bullet c).$$

From $\mathbf{k}, \mathbf{s}$ it is easy to define other common functions, in particular the identity $\mathbf{i} = \mathbf{s}\mathbf{k}\mathbf{k}$, the second projection $\overline{\mathbf{k}} = \mathbf{k}\mathbf{i}$.

A combinatory algebra is then a *sort of* monoid equipped with an equivalence relation generated by a notion of reduction (this hides the computationally-relevant part of this technique!).

Such a structure cannot be associative, as if it were we would have

$$\mathbf{k} = \mathbf{k}\mathbf{k}\mathbf{k} = \mathbf{k}(\mathbf{k}\mathbf{k}), \quad \text{then } a = \mathbf{k}a\mathbf{k} = \mathbf{k}(\mathbf{k}\mathbf{k})a\mathbf{k} = \mathbf{k}\mathbf{k}\mathbf{k} = \mathbf{k},$$

nor commutative, as

$$\mathbf{i} = \mathbf{k}\mathbf{i}(\overline{\mathbf{k}\mathbf{i}}) = \overline{\mathbf{k}\mathbf{i}}(\mathbf{k}\mathbf{i}) = \mathbf{k}\mathbf{i}, \quad \text{then } a = \mathbf{i}a = \mathbf{k}\mathbf{i}a = \mathbf{i}.$$

In both cases, it would be the trivial combinatory algebra.

Formulæ of intuitionistic propositional logic can be canonically associated to elements of a combinatory algebra; this is known as the *Curry-Howard correspondance*.

Kleene's realizability can be seen as a loose version of the correspondance for Heyting's arithmetics.

In 1990, Griffin showed that a well-known program introduced in Scheme—the call-with-current-continuation program (`cc`)—could be associated to excluded middle, thus extending the above correspondance to classical logic.

Starting from this point, Krivine tried to generalise Kleene's technique to classical theories, namely ZF.

Curry (1934) Functionality in Combinatory Logic.

Howard (1980) The formulæ-as-types notion of construction.

Griffin (1990) The Formulæ-as-Types Notion of Control.

Krivine's construction begins with the definition of a new structure, encompassing a computational system (λ -calculus) and the `cc` operator.

A realizability algebra is a tuple $\mathcal{K} = (\Lambda_c, \Pi, \succ, \perp\!\!\!\perp)$ such that

- the set of terms Λ_c is an extension of λ -calculus with `cc`,
- the set of stacks Π is the set of finite sequences of terms, or $\Pi = \Lambda_c^{<\omega}$,
- $\succ$ is a relation on $\Lambda_c \times \Pi$ encompassing β -reduction, and
- $\perp\!\!\!\perp$ is a set of processes (t, π) closed by antireduction, i.e. $(t, \pi) \succ (u, \rho)$ and $(u, \rho) \in \perp\!\!\!\perp$ implies $(t, \pi) \in \perp\!\!\!\perp$.

Krivine (2001) Typed lambda-calculus in classical Zermelo-Fr enkel set theory.

This structure induces an interpretation for formulæ in $\wp(\Pi)$.

For a set $X \subseteq \Pi$, define $X^\perp = \{t \in \Lambda_c \mid \forall \pi \in X ((t, \pi) \in \perp)\}$. This is used in the definition of a binary operation $\cdot \rightarrow \cdot$ on $\wp(\Pi)$ specified as

$$X \rightarrow Y = \{t \cdot \pi \mid t \in X^\perp, \pi \in Y\}.$$

Then, by induction we can define

$$\begin{aligned} \|\perp\| &= \Pi, & \|\top\| &= \emptyset, \\ \|\varphi \rightarrow \psi\| &= \|\varphi\| \rightarrow \|\psi\|, \\ \|\forall x \varphi(x)\| &= \bigcup_{c \in M} \|\varphi[c/x]\|. \end{aligned}$$

A formula φ is realized by $t \in \Lambda_c$ if $(t, \pi) \in \perp\!\!\!\perp$ for all $\pi \in \|\varphi\|$ (i.e. $t \in \|\varphi\|^{\perp\!\!\!\perp}$). In symbols, $t \Vdash \varphi$.

An interpretation for ZF-formulæ can be provided within this formalism, and explicit realizers for axioms can be defined.¹ Then, ZF is a subset of $\{\varphi \mid \exists t(t \Vdash \varphi)\}$.

¹Again in Krivine (2001).

Generally speaking, the set $\{\varphi \mid \exists t(t \Vdash \varphi)\}$ is not a coherent theory.

For this reason, it is necessary to define a subset of terms $Q \subset \Lambda_c$, in order to obtain a coherent theory

$$\mathcal{T} = \{\varphi \mid \exists t \in Q(t \Vdash \varphi)\}$$

(*ça va sans dire*, ZF-realizers are contained in Q).

A realizability model for set theory is any model satisfying $\mathcal{T}$.

The implication $\rightarrow$ induces a preorder on $\wp(\Pi)$, defined by

$$X \leq_{\perp} Y \quad \text{iff} \quad \exists t \in \mathbf{Q}(t \Vdash X \rightarrow Y)$$

The associated poset carries the structure of a boolean algebra.

$\mathcal{T}$ is then obtained from the interpretation of ZF into a boolean algebra.

Recently, different structures have been proposed to unify boolean algebras, pcas, and Krivine's structures.

While forcing can be shown to be a particular case of realizability (the “commutative case”), whether any realizability model can be described as a generic extension is still an open question.

Miquel (2020) Implicative algebras: a new foundation for realizability and forcing

Cohen, Miquey, Tate (2021) Evidenced Frames: A Unifying Framework Broadening Realizability Models

Brieg, van der Berg (2023) Arrow algebras

As in forcing, cardinals may collapse. To prevent this, we developed an analogue for κ -closure.

In the context of complete boolean algebras, κ -closure states that $B \setminus \{\mathbf{0}\}$ contains a dense κ -closed subset.

Equivalently, a complete boolean algebra $\mathbb{B}$ is κ -closed if and only if for any $\gamma < \kappa$ and for any sequence $(b_\xi)_{\xi < \gamma}$ of elements of $B \setminus \{\mathbf{0}\}$, if for any $\alpha < \gamma$, $\bigwedge_{\xi < \alpha} b_\xi \neq \mathbf{0}$, then $\bigwedge_{\xi < \gamma} b_\xi \neq \mathbf{0}$.

To find an analogue of κ -closure for realizability algebras, it is necessary to dispose of terms that represent ordinals below κ .

We then consider a realizability algebra $\mathcal{K} = (\Lambda_{\mathbf{c}}, \Pi, \succ, \perp)$ which contains for any $\alpha < \kappa$, a term $\underline{\alpha}$ such that $(\underline{s})\underline{\alpha} \succ \underline{\alpha + 1}$; furthermore, we assume the existence of a term χ such that

$$\chi \underline{\alpha} \underline{\beta} t u \succ \begin{cases} t & \text{if } \alpha < \beta; \\ u & \text{otherwise} \end{cases}$$

for any $\alpha, \beta < \kappa$, $t, u \in \Lambda_{\mathbf{c}}$.

In addition to that, we also suppose the existence of a sequence $(u_{\xi})_{\xi < \kappa}$ such that, for any $\alpha < \gamma < \kappa$, $(\underline{\gamma})\underline{\alpha} \succ u_{\alpha}$.

Terms of the form $\underline{\alpha}$ are needed to control ordinals' representatives $\hat{\alpha}$ in the expected way.

LEMMA

For any $\alpha \leq \kappa$ there exists a term such that $t \Vdash \text{Ord}(\hat{\alpha})$.

Furthermore, these representatives are computationally well-behaved.

We will say that a term t *accumulates* a sequence $(u_\xi)_{\xi < \gamma}$ whenever $(t)\underline{\alpha} \gg u_\alpha$ for any $\alpha < \gamma$.

A realizability algebra is κ -closed for some cardinal κ if for any $\gamma < \kappa$ and for every sequence $(u_\xi)_{\xi < \gamma} \subseteq \Lambda_c$

- there is a term t that accumulates $(u_\xi)_{\xi < \gamma}$;
- for any term v , if for any $\delta < \gamma$ there exists an accumulator w of the sequence $(u_\xi)_{\xi < \delta}$ such that $v\{x := w\} \not\Vdash \perp$, then for any accumulator t of the sequence $(u_\xi)_{\xi < \gamma}$, $v\{x := t\} \not\Vdash \perp$.

That is to say, “any $< \kappa$ -sequence of terms can be recollected into a term t which preserves the compatibility of the sequence”.

This property corresponds to κ -closure in forcing!

From a boolean algebra $\mathbb{B}$, we can build a realizability algebra $\mathcal{K}_{\mathbb{B}}$ that generates the same model.

PROPOSITION [FONTANELLA, F.]

For any cardinal κ , $\mathbb{B}$ is κ -closed if and only if $\mathcal{K}_{\mathbb{B}}$ is κ -closed.

For the translation boolean algebras/realizability algebras
Matthews, A guide to Krivine realizability for set theory (2023).

A generalized version of results from Krivine (and Berardi Bezem Coquand) about choice and continuum hypothesis can be obtained.

THEOREM [FONTANELLA, F.]

Given a κ -closed realizability algebra $\mathcal{K}$, the realizability model N satisfies $\text{AC}_{<\hat{\kappa}}$.

THEOREM [FONTANELLA, F.]

Given a κ -closed realizability algebra $\mathcal{K}$, if $M \models \text{GCH}$, then $N \models \text{CH}_{<\hat{\kappa}}$.

Berardi Bezem Coquand (1998) On the computational content of the axiom of choice.

Krivine (2016) Bar recursion in classical realizability.

We obtain an explicit realizer for the preservation of cardinals.

THEOREM [FONTANELLA, F.]

Given a κ -closed realizability algebra $\mathcal{K}$, there exists a term such that for any cardinal $\mu \leq \kappa$ and for any ordinal $\gamma < \mu$, it is realized that no surjection exists from $\hat{\gamma}$ onto $\hat{\mu}$.

Thank you for the attention